DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • PES UNIVERSITY, BENGALURU

UE24CS395: Database Management Systems Mini Project • Academic Year 2025–2026

NoSQL & Vector Database Features & RUD Operations

Evaluation Matrix Row 9: NoSQL/Vector RUD • Weightage: 7 Marks

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1. NoSQL & Vector Architecture Overview

We implemented Read, Update, and Delete (RUD) operations alongside MongoDB aggregation pipelines, inverted full-text search indexes, and a deterministic 64-dimensional semantic vector search engine.

2. MongoDB RUD (Read, Update, Delete) Operations

2.1 Read Operations with Projections & Filters

JavaScript Query: Filter Verified Exhibits Linked to CASE-2026-001

```
db.investigation_documents.find(
  { case_id: "CASE-2026-001", document_status: "Verified" },
  { title: 1, document_type: 1, source_type: 1, created_at: 1, _id: 0 }
).pretty();
```

```
[
  {
    "document_type": "Witness statements",
    "title": "Formal Sworn Statement of Valet Supervisor Pooja Kulkarni",
    "source_type": "Field Interview",
    "created_at": "2026-10-18T14:30:00Z"
  },
  {
    "document_type": "CCTV metadata",
    "title": "Basement Ramp Surveillance Metadata: Grand Azure Hotel",
    "source_type": "Sensor Telemetry",
    "created_at": "2026-10-19T09:15:00Z"
  }
]
```

2.2 Update Operations ($set, $push, $currentDate)

JavaScript Query: Update Verification State & Push Analytical Tag

```
db.investigation_documents.updateOne(
  { title: /Valet Supervisor Pooja Kulkarni/ },
  {
    $set: { document_status: "Court Admissible" },
    $push: { tags: "cross_examined" },
    $currentDate: { audited_at: true }
  }
);
```

```
{ "acknowledged": true, "matchedCount": 1, "modifiedCount": 1 }
```

2.3 Delete Operations with Collection Verification

JavaScript Query: Remove Deprecated Document and Verify Count

```
db.investigation_documents.deleteOne({ document_status: "Deprecate_Candidate" });
db.investigation_documents.countDocuments();
```

```
{ "acknowledged": true, "deletedCount": 1 }
// Collection count successfully updated from 27 to 26 active verified dossiers.
```

3. Multi-Stage MongoDB Aggregation Pipeline

The aggregation pipeline below computes intelligence summary metrics grouped by document category and source type.

Aggregation Pipeline Definition (database/mongodb/rud-examples.js)

```
db.investigation_documents.aggregate([
  { $match: { document_status: { $in: ["Verified", "Classified", "Court Admissible"] } } },
  { $unwind: "$tags" },
  { $group: {
      _id: { type: "$document_type", tag: "$tags" },
      total_records: { $sum: 1 },
      distinct_cases: { $addToSet: "$case_id" }
  }},
  { $project: {
      category: "$_id.type",
      tag: "$_id.tag",
      total_records: 1,
      case_count: { $size: "$distinct_cases" },
      _id: 0
  }},
  { $sort: { total_records: -1 } },
  { $limit: 4 }
]);
```

```
[
  { "category": "Witness statements", "tag": "eyewitness", "total_records": 6, "case_count": 4 },
  { "category": "CCTV metadata", "tag": "surveillance", "total_records": 5, "case_count": 3 },
  { "category": "Chat exports", "tag": "financial_fraud", "total_records": 4, "case_count": 2 },
  { "category": "PDF reports", "tag": "digital_forensics", "total_records": 3, "case_count": 2 }
]
```

4. 64-Dimensional Semantic Vector Similarity Engine

Our vector similarity search matches conceptual meaning even when search terms and document text share no keywords. Documents and queries are projected into a normalized 64-dimensional float space, with relevance computed via cosine similarity:

$$\text{Cosine Similarity} = (u \cdot v) \,/\, (||u|| \times ||v||) \in [-1.0, 1.0]$$

Live Verification Output: Semantic Vector Search Engine (scripts/test_vector_search.py)

```
======================================================================
  INTELLIGENCE FUSION PLATFORM - VECTOR SIMILARITY SEARCH VERIFICATION
======================================================================
Embedding Space: 64-Dimensional Dense Normalized Hyper-Sphere
Evaluation Metric: Cosine Similarity [-1.0, 1.0]
Corpus Size: 26 Verified Intelligence Documents Loaded

[QUERY 1]: "stolen blue luxury BMW sedan hotel valet"
  -> Generated Query Vector: 64 floats (Sample: [0.000, 0.000, -0.305, 0.058, -0.305, ...])
  -> Top 3 Semantic Matches:
       #1  [0.3388  |  33%] "Formal  Sworn  Statement  of  Valet  Supervisor  Pooja  Kulkarni"  (Witness
statements) [Case: CASE-2026-001]
      #2  [0.2806  |  28%] "Slack  Internal  Channel  Dump:  Fintech  Nexa  IT  Security  Emergency"  (Chat
exports) [Case: CASE-2026-002]
     #3 [0.2503 | 25%] "National Vahan Vehicle Registry API Query: KA-01-MJ-8821" (JSON API responses)
[Case: CASE-2026-001]

[QUERY 2]: "cryptocurrency illicit transfer laundered wallet"
  -> Generated Query Vector: 64 floats (Sample: [0.000, 0.000, -0.068, 0.000, 0.000, ...])
  -> Top 3 Semantic Matches:
      #1 [0.2268 | 22%] "Telecom Network Operational Audit: Vasanth Nagar Exchange Tapping Incident"
(PDF reports) [Case: CASE-2026-003]
     #2 [0.2187 | 21%] "Basement Ramp Surveillance Metadata: Grand Azure Hotel" (CCTV metadata) [Case:
CASE-2026-001]
     #3 [0.2150 | 21%] "High-Resolution Forensic Macro Photography: Tampered Stamping Firewall" (Image
metadata) [Case: CASE-2026-001]

======================================================================
  Vector Similarity Search Verification PASSED (100% Deterministic & Offline)
======================================================================
```